

IMU x, y opposed to the optical body

1 Definitions

Denote with “optical body” the rigid body tracked by Vicon. Its position in the world frame of reference is $\mathbf{p}$ and its attitude is the quaternion q . The quaternion encodes a rotation $R(q)$ from body to world reference frames

$$\mathbf{v}_{\text{world}} = R(q) \mathbf{v}_{\text{body}}. \quad (1)$$

The z coordinate in the optical body frame is oriented upwards.

The IMU is expressed in a second frame of reference on the same object.

2 Problem

The IMU frame of reference was assumed to coincide with the “optical” frame of reference. However, it was observed that while the IMU and optical z axes match on all sequences, the x and y axes seem to point in opposite directions. On these sequences, this inversion can be fixed by a rotation of π radians around the z axis:

$$R_z(\pi) = \text{diag}(-1, -1, +1), \quad \mathbf{u}_{\text{body}} = R_z(\pi) \mathbf{u}_{\text{imu}}. \quad (2)$$

3 Diagnostics Tests

In this section we describe the tests used to uncover the problem. We apply the tests to the first two minutes of 7 recordings.

3.1 Angular velocity test

Denote $\boldsymbol{\omega}_k$ the k -th angular velocity sample, and with $\Delta t = 0.01$ s the sampling period.

Over one sample the attitude advances by a body rotation,

$$q_{k+1} = q_k \otimes \exp(\boldsymbol{\omega}_k \Delta t),$$

where $\exp$ denotes the exponential map that converts the rotation vector $\boldsymbol{\omega}_k \Delta t$ to a corresponding unit quaternion. Left-multiplying by q_k^{-1} we isolate the increment,

$$q_k^{-1} \otimes q_{k+1} = \exp(\boldsymbol{\omega}_k \Delta t).$$

Inverting with the logarithmic map for a rotation of angle less than π , we obtain the **angular velocity in the optical body frame** obtained from vicon:

$$\boldsymbol{\omega}_k = \frac{1}{\Delta t} \text{Log}(q_k^{-1} \otimes q_{k+1}). \quad (3)$$

We want to compare this with IMU angular velocities in the IMU body frame, $\boldsymbol{\omega}_k^{\text{imu}}$.

First, we smooth the IMU measurements with a local average, $\tilde{\boldsymbol{\omega}}_k = \frac{1}{2}(\boldsymbol{\omega}_k^{\text{imu}} + \boldsymbol{\omega}_{k+1}^{\text{imu}})$. Further, we consider only those samples with a large enough velocity, $\|\tilde{\boldsymbol{\omega}}\| > 0.5$ rad/s. Then, we compute the per-axis correlations

$$\mathbf{c} = \text{corr}_k(\tilde{\boldsymbol{\omega}}_k, \boldsymbol{\omega}_k) \quad (4)$$

Large and negative correlations in the x and y axes indicate that these axes are inverted. Moreover, this can be used as a criterion to decide when to apply the correction (2) to reverse the axes:

$$\text{reverse } x, y \iff -c_x - c_y + c_z > c_x + c_y + c_z \iff c_x + c_y < 0. \quad (5)$$

Figure 1 shows a sequence where the IMU and optical frames agree: the two angular velocities $\tilde{\omega}$ and ω overlap. On the contrary, figure 2 shows a case where both frames differ: the x and y coordinates of both angular velocities clearly differ, while the z coordinate remains the same.

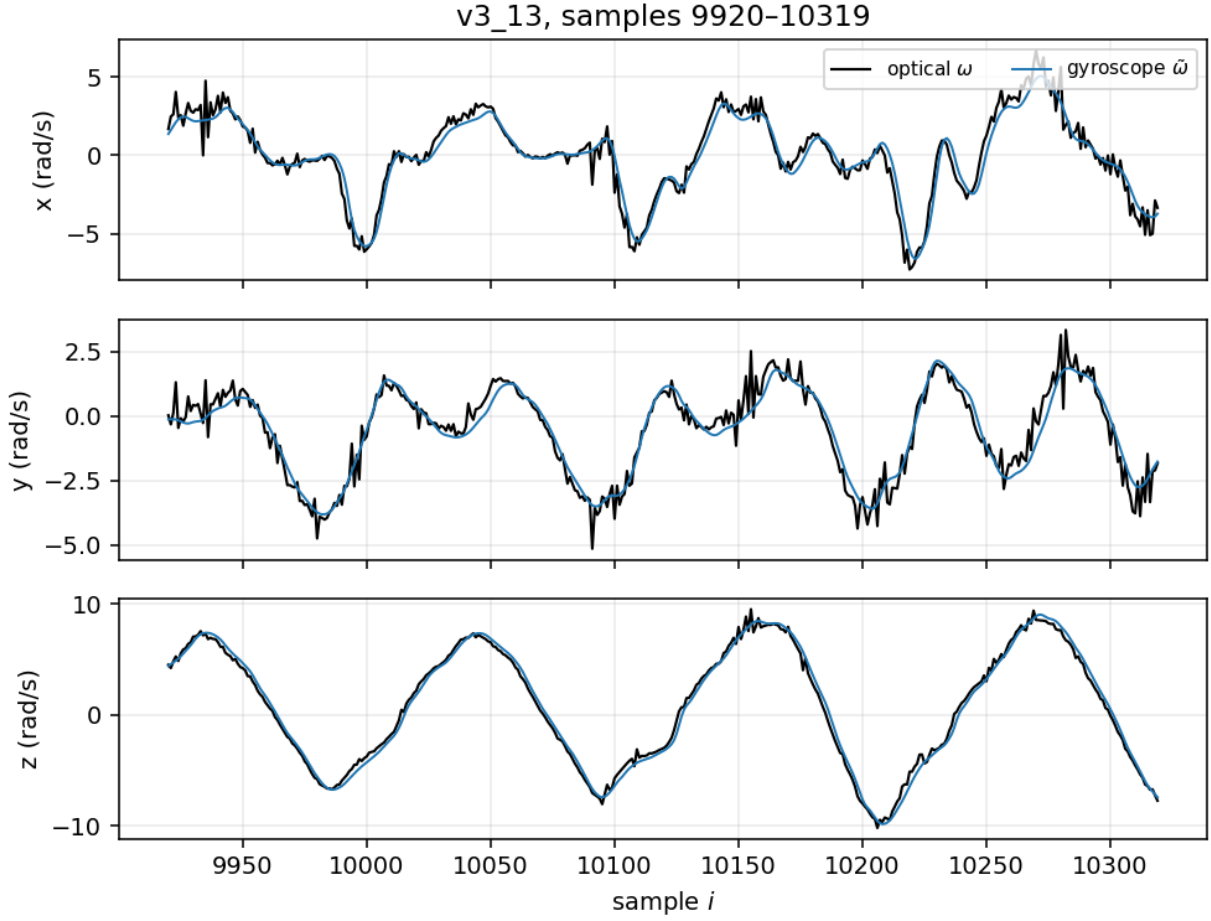


Figure 1: Optical rate ω_i and gyroscope midpoint $\tilde{\omega}_i$ on v3_13, whose axes agree. The two traces overlap.

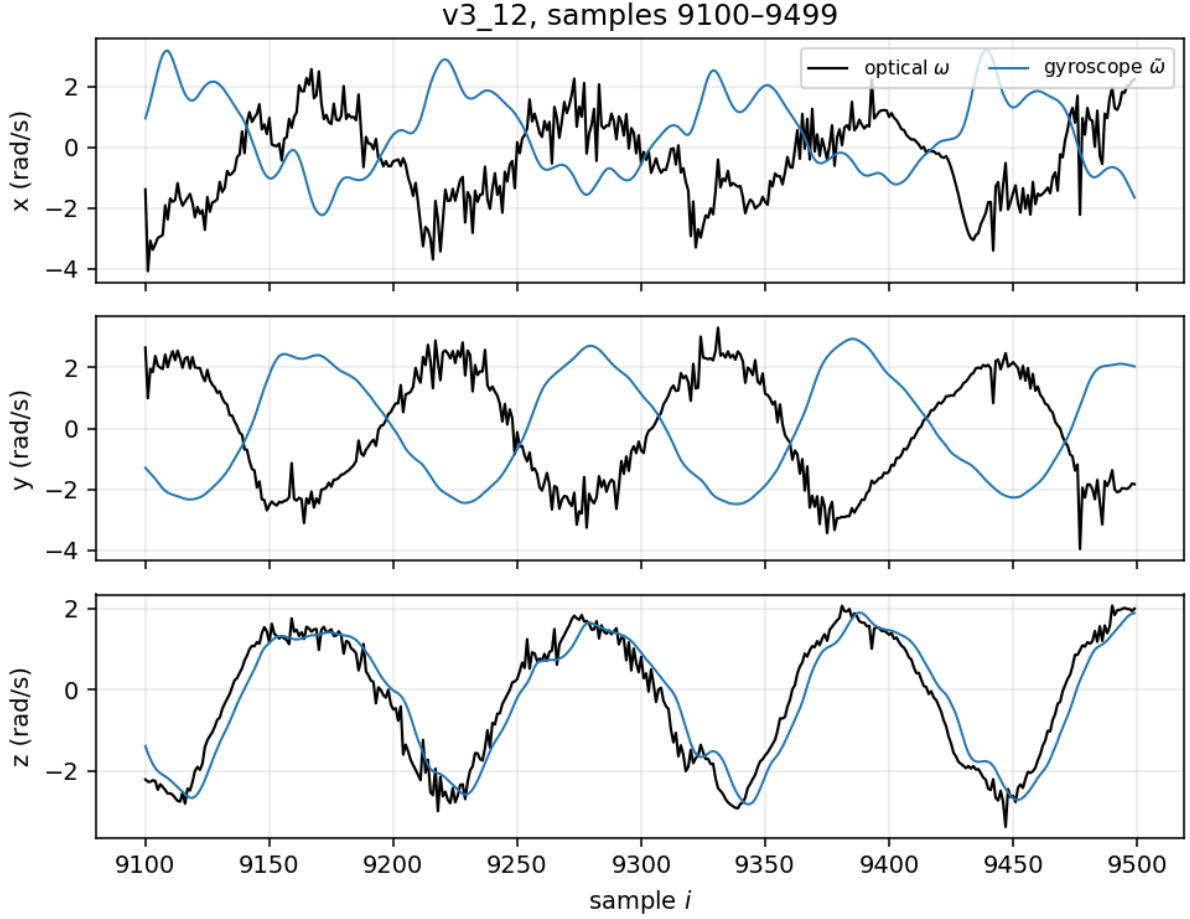


Figure 2: The same comparison on v3_12. The shapes match, with x and y inverted and z unchanged.

3.2 Acceleration test

First, we compute the “optical specific force” $\mathbf{f}$ as the second-derivative of the optical position $\mathbf{p}$

$$\mathbf{f}_k = \frac{\mathbf{p}_{k+1} - 2\mathbf{p}_k + \mathbf{p}_{k-1}}{\Delta t^2} + \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}. \quad (6)$$

This specific force is expressed in the world frame of reference.

Then, we express the IMU acceleration $\mathbf{a}^{\text{imu}}$ in the world frame of reference. This implies rotating with the correction matrix R_z from (2), if needed, and then with the rotation matrix $R(q)$. We write the world-frame acceleration as

$$\mathbf{a}^{\text{world}} = R(q) R \mathbf{a}^{\text{imu}}. \quad R \in \{I, R_z(\pi)\}. \quad (7)$$

Figures 3 and 4 show the behavior of the specific force when using $R = I$. For an aligned sequence, Figure 3 shows that both traces overlap, while Figure 4 shows that the traces differ on an unaligned sequence: the vertical component of the accelerometer sits near $-g$ while f_z sits near $+g$, while the other two components are not mirrored due to the effect of $R(q)$.

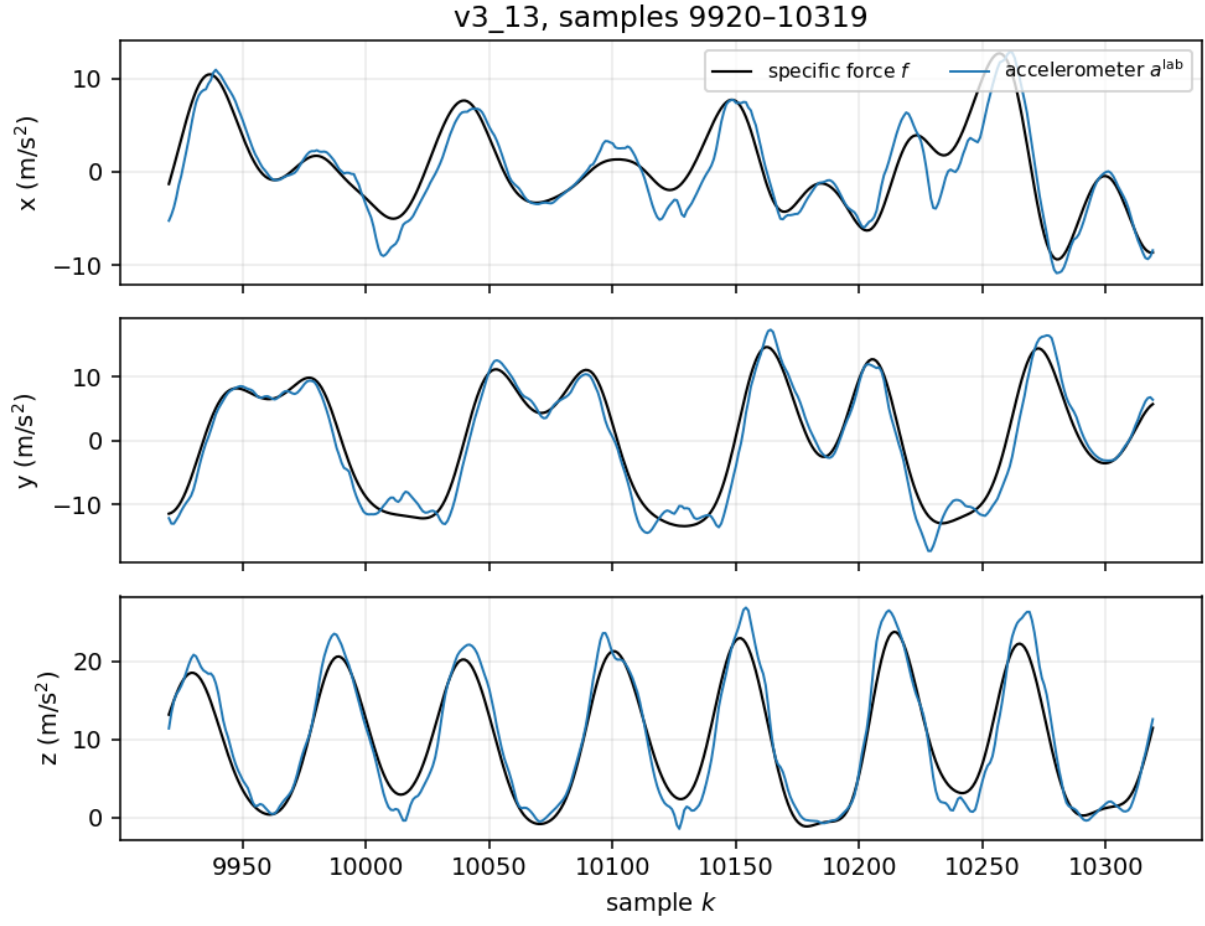


Figure 3: Specific force f_k and lab accelerometer a_k^{world} at $R = I$ on v3_13, the same stretch as Figure 1. The accelerometer follows f .

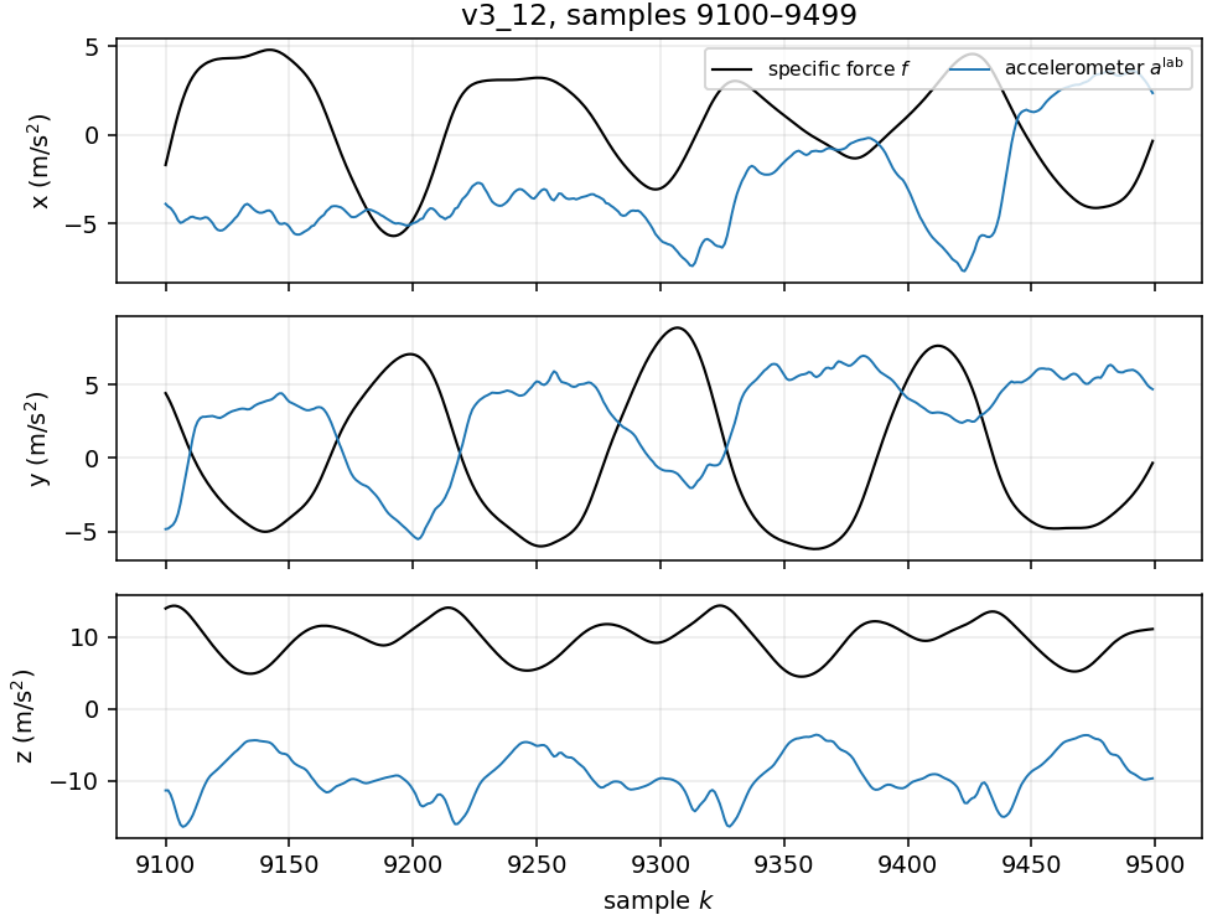


Figure 4: The same comparison on v3_12, the stretch of Figure 2. The traces run against each other, and the lab vertical of the accelerometer sits near $-g$ while f_z sits near $+g$.

Once more, a numerical test to determine the need to correct can be based on correlations. First compute the correlation coefficients between both measurements, and then average them over all components:

$$\begin{aligned}\gamma &= \text{corr}(\mathbf{a}^{\text{world}}, \mathbf{f}) \\ \Gamma &= \frac{1}{3} \sum_{i=1}^3 \gamma_i.\end{aligned}\tag{8}$$

The correlation Γ can be computed in the two cases, $R = I$ and $R = R_z$. If the correlation for the second case is larger than for the first one, then the correction is needed.

3.3 Numerical comparisons

The table below shows the numerical indicators that can be used to determine the need to apply the correction (2). For each test sequence, the component-wise gyroscope correlations (4) and the accelerometer correlations (8) are shown.

sequence	gyroscope, (3)			accelerometer, (7)	
	c_x	c_y	c_z	$R = I$	$R = R_z(\pi)$
v3_01	+0.51	+0.45	+0.88	+0.89	−0.79
v3_02	−0.30	−0.35	+0.72	−0.48	+0.86
v3_11	+0.45	+0.60	+0.78	+0.65	−0.47
v3_12	−0.60	−0.70	+0.87	−0.57	+0.95
v3_13	+0.95	+0.77	+0.99	+0.96	−0.91
v3_14	−0.46	−0.69	+0.71	−0.34	+0.80
v3_15	−0.45	−0.55	+0.74	−0.44	+0.92

It can be seen that in all cases both criteria agree. Sequences v3_01, v3_11 and v3_13 don't need to be corrected, while sequences v3_02, v3_12, v3_14 and v3_15 need to be corrected.

Note that c_z is positive on all seven sequences (+0.71 to +0.99), so the vertical axes agree.

4 Roll and pitch from the accelerometer

At rest the accelerometer reads the specific force of gravity in the body frame,

$$\mathbf{a} = R(q)^\top \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}.$$

Write the optical attitude as the intrinsic Z-Y-X rotation $R(q) = R_z(\psi) R_y(\theta) R_x(\phi)$, with roll ϕ , pitch θ and yaw ψ . A yaw about the lab vertical leaves $(0, 0, g)^\top$ unchanged, so ψ drops out and

$$\begin{aligned} a_x &= -g \sin \theta, \\ a_y &= g \cos \theta \sin \phi, \\ a_z &= g \cos \theta \cos \phi. \end{aligned}$$

For $\theta \in [-\pi/2, \pi/2]$ one has $\cos \theta \geq 0$, and the two angles are

$$\phi = \text{atan2}(a_y, a_z), \quad \theta = \text{atan2}(-a_x, \sqrt{a_y^2 + a_z^2}). \quad (9)$$

Replacing (a_x, a_y) by $(-a_x, -a_y)$, which is what $R_z(\pi)$ does, changes (ϕ, θ) to $(-\phi, -\theta)$.

Note that the formulas hold when $\mathbf{a}$ is gravity. We can consider them approximate under mild accelerations, but they will become invalid during hard motion. Since roll is read from the two components a_y and a_z , it is estimated from a projection of gravity with weight $\cos \theta$. When the optical pitch is large, that projection shrinks while the linear acceleration does not, so the accelerometer roll departs from the optical roll, making it unusable. On the contrary, pitch is read from $a_x = -g \sin \theta$ which holds most of g when the pitch grows, so it continues to follow the optical pitch. For this reason, in analysis below we focus only on pitch.

We evaluated these pitch estimate from the accelerometer, and compared it with pitch of the optical quaternion. We focus on the first 10s of data, where $\|\mathbf{a}\|$ is closest to $\mathbf{g}$.

Figure 5 shows that both estimates match when the sequence doesn't require inversion, while otherwise they become inverted as well.

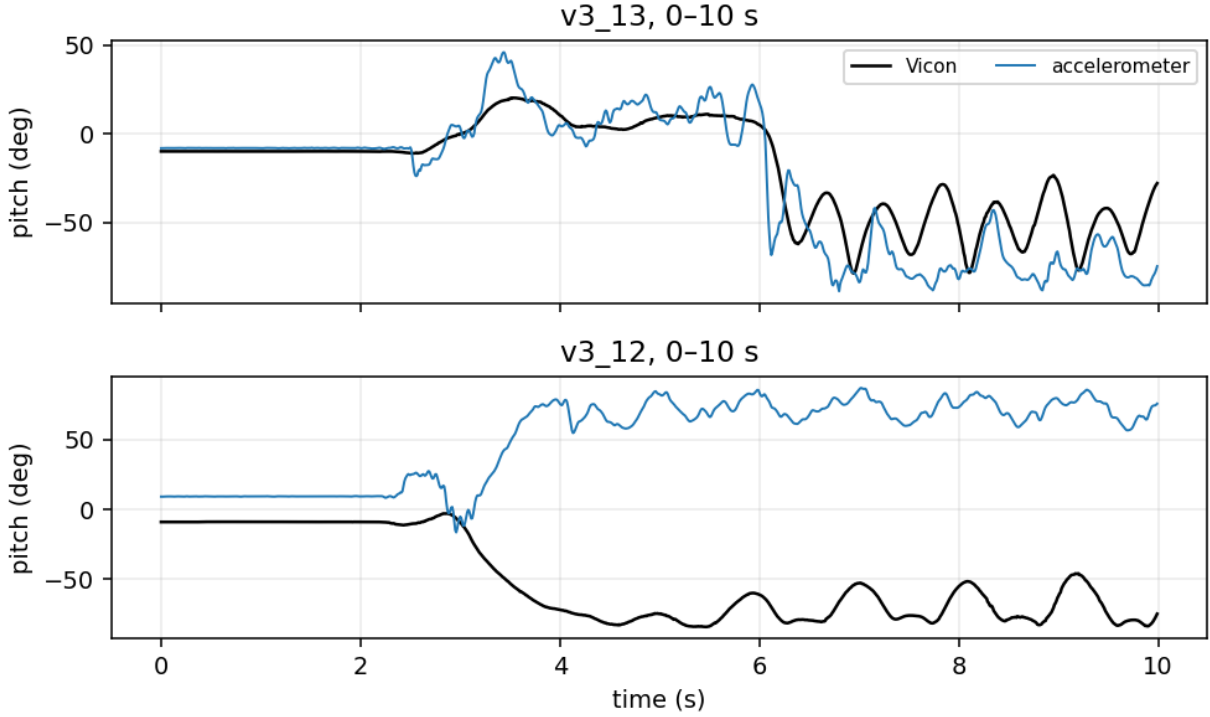


Figure 5: Pitch from (9) and from the optical quaternion, over the first 10s. On v3_13 the axes agree and the two pitches follow each other. On v3_12 the horizontal axes are reversed, and the accelerometer pitch is the negative of the optical pitch.

Table 1 shows the RMSE of the pitch difference, in degrees, over the first 120s, with and without applying the correction from (5). It can be seen that the correction realigns the pitches, bringing their RMSE down to a similar level for all sequences.

Table 1: RMSE between optical pitch and accelerometer estimate from (9), with and without correction.

sequence	Not corrected	Corrected
v3_01	26.5	26.5
v3_02	139.2	18.8
v3_11	43.9	43.9
v3_12	139.7	19.5
v3_13	29.8	29.8
v3_14	142.1	12.5
v3_15	126.6	18.9

5 Correction

When (5) holds, the measurements are mapped into the optical body and the pose is left as recorded:

$$\mathbf{a} \leftarrow R_z(\pi) \mathbf{a}, \quad \boldsymbol{\omega} \leftarrow R_z(\pi) \boldsymbol{\omega}, \quad q \text{ and } \mathbf{p} \text{ unchanged.} \quad (10)$$

6 Effect of the correction

In this section, we test the effect of the proposed test and correction on an actual attitude estimator. We apply the same estimator twice, once with the axes as recorded and once after (10). We choose as estimator VQF, in 6-axis mode with $\tau_{\text{acc}} = 3$.

The error quaternion is $\hat{q} \otimes q^{-1}$. With R the rotation matrix of that quaternion, the error angle is

$$\theta = \arccos \frac{\text{tr}(R) - 1}{2}. \quad (11)$$

Table 2 shows the RMSE of θ , in degrees, over the first 120 s, with and without applying the correction from (5). It can be seen how the correction drastically reduces the error in the affected sequences.

Table 2: RMSE of the rotation angle θ , with and without correction.

sequence	Not corrected	Corrected
v3_01	23.3	23.3
v3_02	143.5	18.6
v3_11	27.5	27.5
v3_12	145.0	20.3
v3_13	27.6	27.6
v3_14	143.4	9.4
v3_15	132.5	10.4